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Prompt γ-ray activation analysis of Martian analogues at the FRM II neutron reactor and the verification of a Monte Carlo planetary radiation environment model

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ABSTRACT

Planetary radiation environment modelling is important to assess the habitability of a planetary body. It is also useful when interpreting the γ-ray data produced by natural emissions from radioisotopes or prompt γ-ray activation analysis. γ-ray spectra acquired in orbit or in-situ by a suitable detector can be converted into meaningful estimates of the concentration of certain elements on the surface of a planet. This paper describes the verification of a Monte Carlo model developed using the MCNPX code at the University of Leicester. The model predicts the performance of a geophysical package containing a γ-ray spectrometer operating at a depth of up to 5 m. The experimental verification of the Monte Carlo model was performed at the FRM II facility in Munich, Germany. The paper demonstrates that the model is in good agreement with the experimental data and can be used to model the performance of an in-situ γ-ray spectrometer.

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1. Introduction

1.1. Prompt gamma activation analysis (PGAA)

Non-nuclear compositional analysis techniques such as X-ray fluorescence spectroscopy (XRF) or X-ray diffraction (XRD) can be destructive. In many cases rock samples have to be sliced or crushed in order to produce a homogenous sample that represents the composition of the whole rock sample. The primary reason for preparing rock samples in this way for X-ray analysis is that X-rays are not very penetrating. In silica of density 2.33 g cm^{-3}, 20 keV X-rays have an interaction length of the order of millimetres (mm); 6 keV X-rays have an interaction length of the order of microns (μm). This is undesirable when examining objects of importance such as meteorites and items of archaeological significance. From a planetary exploration point of view, sample preparation can be difficult and an analysis technique that is non-destructive and enables the analysis of the bulk of a planetary regolith can be useful.

Prompt gamma activation analysis is a nuclear analytical technique that can be used to determine the composition of a sample by using the γ-rays that are emitted as a result of neutron irradiation. Neutrons can interact with a sample via inelastic scattering, non-elastic scattering or neutron capture reactions. PGAA is usually carried out using thermal neutron beams, which favour neutron capture reactions; whereas γ-ray spectroscopy (GRS) examines the γ-ray emissions that result from all of these reactions [1]. PGAA leaves little to no residual radioactivity in the sample, so an object analysed in this way can be reused and stored without any special requirements.

There are many advantages of using PGAA as a compositional analysis technique in comparison with XRF or XRD (techniques that can be used in the laboratory as well as in in-situ planetary missions):

- Neutrons are more penetrating than X-rays, allowing in-depth analysis of the samples of interest. To examine the unaltered regions of a planetary regolith, it is necessary to probe at depths >2 m; or in the case of a rover to find a suitable outcrop which is not always possible. X-rays cannot penetrate very deep into a surface (μm to mm); therefore X-ray techniques would rely on some form of sample retrieval system. γ-rays can penetrate several metres of matter, depending on density and γ-ray energy, therefore a detector could sit on the surface and collect data on these depths without the need for any retrieval systems.

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- γ-rays are more penetrating than X-rays; therefore the technique does not need to be carried out in a vacuum.
- PGAA is a non-destructive method of compositional analysis, resulting in the average bulk composition of an object without any sample preparation; XRF requires the sample to be crushed into a homogenous powder to get an accurate estimate of the average bulk composition. This requires (in addition to a sample retrieval system) a preparation system.
- PGAA can be used to identify several elements that can be difficult to detect with XRF, including low *Z* elements such as B, H, C, N, P and also toxic elements such as Cd and Pb [1].
- The sample can be a solid, liquid or gas.
- PGAA detector systems are designed to be robust. γ-rays have higher energies than X-rays; therefore they have longer interaction lengths. It is possible for a PGAA detector system to carry out PGAA through the walls of a probe. In the case of XRF or XRD a window is required otherwise the X-rays would be absorbed by the probe walls. PGAA has been applied in the oil and well logging industry where high pressure and temperature conditions require the detector to be robust [2].

There are a few disadvantages of using PGAA (and GRS) compared to XRF and XRD on remote sensing missions:

- The detectors required for PGAA and GRS need larger volumes than those used in XRF and XRD because γ-rays have a larger interaction length compared to X-rays. This can increase the mass of the instrument; which could be undesirable in remote sensing missions where mass is limited.
- Current remote sensing GRS missions rely on natural γ-ray emissions from the planetary surface (see Section 1.2). Long time periods are necessary to get γ-ray spectra with sufficient counting statistics. In contrast XRF and XRD instruments include an X-ray source to excite X-ray emission, that significantly reduces the time required to obtain good counting statistics. A neutron source could be used to induce additional γ-ray emission from a surface; however, these have yet to be used in a remote sensing or in-situ GRS mission.
- XRD is a well-established technique to image a planetary surface [3,4] in addition to determining the composition of an object, whereas γ-ray imaging has not been carried out on a planetary surface to date.

These advantages and disadvantages demonstrate that XRF, XRD and PGAA/GRS are complementary techniques; ideally all techniques should be used to fully characterise a planet's surface.

The possible terrestrial uses of PGAA are extensive and include; screening of cargo at airports for explosives and other sensitive locations, food contamination analysis, the analysis of the composition of fossil fuels and the purity of materials; the analysis of the environmental impact of pollutants and the non-destructive analysis of archaeological samples [5]. These applications require neutron sources to induce PGAA.

1.2. Planetary γ-ray emission

When galactic and solar cosmic rays (GCR and SCRs) directly interact with a planetary atmosphere or surface, a complex series of interactions occur (including spallation, ionisation and absorption; these and other reactions are detailed in Knoll [6]), producing secondary particles such as neutrons and high-energy γ-rays (see Fig. 1). γ-ray emission may also occur as a result of the decay of radioactive nuclei, such as U, Th and K. The processes that generate γ-ray emission (including radioisotope activation, inelastic scattering, neutron capture reactions and radioactive decay) from a planetary surface are summarised in Boynton et al. [7]. A γ-ray detector in orbit around or on the surface of a planet can be used to determine the bulk composition of a planetary body. γ-ray spectrometers have been successfully used in remote sensing instruments on-board Lunar Prospector (LP) [8] and 2001 Mars Odyssey (MO), respectively [7]. For in-situ GRS, a neutron source (such as ^{252}Cf) may be used to increase the γ-ray emission from the surface [9,10].

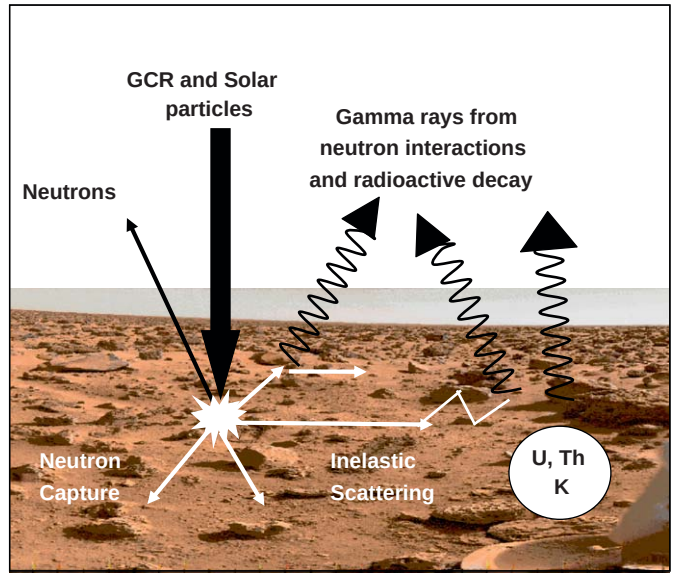

Fig. 1. How γ-ray emission is generated on a planetary surface.

1.3. PGAA on Martian analogues—simulating a planetary surface

A research team led by R. Ambrosi at the University of Leicester is currently developing instrumentation for geophysical applications. The instrumentation includes a geophysical package for a future in-situ planetary science missions that can simultaneously carry out GRS, γ-ray densitometry and radiometric dating of a planetary surface. The geophysical package could be installed on a lander or (ideally) a sub-surface probe that would carry out measurements at depth. This paper focuses on the GRS applications of the detector. The advantages, goals and challenges of a sub-surface GRS mission are detailed in Skidmore et al. [11]. The current design of the geophysical package includes a compact neutron source [9,10], several Lanthanum Bromide ($LaBr_3$(Ce)) scintillator detectors, which are compact, radiation tolerant, operate at a range of temperatures and low in mass [11].

A planetary radiation environment model is being developed with the Monte Carlo N-particle transport code (MCNPX) [12]. MCNPX is a general purpose Monte Carlo radiation transport code developed by Los Alamos National Laboratory that is capable of modelling coupled particle transport through matter. MCNPX can model the physics involved when GCR and SCR interact with matter, from the intranuclear cascade that results in neutron generation to the eventual photon (γ-ray) production. It has previously been used to model the interaction of GCRs with planetary bodies including the Moon [13]. The model can be

adapted to any planetary surface and will be used to model the performance of the geophysical package on a range of planetary surfaces. In the laboratory, the sources used for GRS have known fluxes and energies, and the composition of a sample can be determined through calibrating the detector with existing samples of known composition. When these techniques are being carried out remotely on or in orbit of another planet this is not possible. Instead a Monte Carlo model is traditionally used to convert the GRS data into concentration estimates of each element on the planetary surface. The interactions are modelled through a surface with the estimated composition of the planetary surface of interest. The modelled spectrum is compared with the spectrum gathered by the remote GRS. An iterative process follows; the composition of the modelled surface is altered until the modelled spectrum matches the spectrum of the GRS. The final modelled composition becomes an accurate estimate of the composition of the planetary surface. Forward models have been used to interpret γ-ray data obtained by the γ-ray spectrometers on MO [7] and LP [8].

This paper describes an experimental campaign designed to verify this model by carrying out PGAA on seven samples of known composition. The model currently has a composition similar to the Martian surface for the purposes of the verification, but it may be altered to model other planetary bodies. Ideally this model should be verified using meteorite samples and data acquired from interplanetary missions; however, Martian meteorites are rare on Earth and are very difficult to obtain, especially in quantities of grams or more. Instead terrestrial analogues of Martian meteorites are used to verify the model. This is useful because it is difficult to obtain enough Martian meteorite material due to the rarity of these samples on Earth ($\sim$60 out of $\sim$35,000 meteorites are from Mars). Analogue material of meteorites can be replaced and therefore composition can be found using existing techniques such as XRF, mass spectroscopy, chemical and gas analysis methods. A description of these techniques can be found in Dyar et al. [14] and Kring et al. [15]. The compositional data is then fed into a model to simulate the γ-ray emission, and then the experimentally obtained PGAA data is used to verify the model. The model can be scaled up to a representative size of a planetary body of interest and because it is a three-dimensional model the geophysical package can be modelled within the simulation. The model can incorporate any detector with a known response function. The response functions of several types of γ-ray detector were calculated in Skidmore et al. [11]. An accurate model could vastly improve the quality of the interpretation of any measurements the geophysical package might make on a planetary surface.

This first part of this paper summarises the experimental campaign carried out in January 2008 in the Advanced Neutron Tomography and Radiography System (ANTARES) facility at the Forschungs-Neutronenquelle Heinz Maier-Leibnitz (FRM II) neutron reactor in Munich to irradiate seven samples of known composition and examine their γ-ray emission. The second part discusses the verification of the MCNPX model using the acquired data.

2. Forschungs-Neutronenquelle Heinz Maier-Leibnitz

The FRM II reactor facility in Munich, Germany consists of a high flux neutron source that relies on the fission of a compact enriched uranium silicide core contained in a D_2O tank [16]. The types of research carried out at FRM II include: condensed matter research, nuclear and applied physics research and the irradiation of samples for medical and neutron activation analysis techniques.

2.1. The advanced neutron tomography and radiography system

The ANTARES facility at FRM II is used for thermal neutron radiography and activation studies. The beam line includes filters and apertures to alter the intensity of the beam and to select parts of the neutron spectrum. The facility is described in detail in Lorenz et al. [17]. The multi-filter consists of four crystals (sapphire, single bismuth crystal, polycrystalline bismuth crystal and beryllium) mounted in a wheel of polyethylene. Lorenz et al. [17] studied the effects of the filters on the neutron beam intensity and published the data in 2006. This data was used to produce an input spectrum for the MCNPX model.

3. Martian analogues

Meteorites that have Martian origins provide a way for scientists to determine the composition of the surface of Mars in more detail than can be achieved by remote methods. In a remote sensing mission, the number and type of analysis techniques used to determine surface composition depend on the mass and power constraints of the remote sensing platform; this is not the case on Earth where more sophisticated chemical and spectroscopic techniques may be carried out [14]. Conclusions on the formation and evolution of meteorites are frequently made by comparing them to analogues found on Earth. For example a Martian meteorite with a high Fe content is often compared to an analogue from a volcanic region of the Earth that is also mafic to draw the conclusion that the meteorite may have been formed in a volcanic region of Mars.

Two terrestrial analogue rocks were obtained from the Smithsonian Institution National Museum of Natural History and one was supplied by one of the authors (GKB) for this study. They were:

- Theo's flow (TF), clinopyroxenite, Ontario, Canada (NMNH 117255-17, Division of Petrology and Volcanology, Department of Mineral Sciences, Smithsonian Institution).
- Colombian River basalt (CRB), Basalt, USA (NMNH 116685, Division of Petrology and Volcanology, Department of Mineral Sciences, Smithsonian Institution).
- Mount Erebus (ME), basalt, Ross Island, Antarctica (supplied by GKB).

As the surface of Mars is dominated by basalt (a rock common on all terrestrial planets), we chose to look at two different basalts from Earth. Basalt can have a variety of compositions and the two basalts used in this study, although not exact compositional analogues to Martian meteorites, do sample two different types of basalt. The Theo's flow analogue is a documented chemical analogue to the Martian meteorite, Nakhla [18].

Two 5 g samples of each rock were sent to the University of Leicester Geology department for XRF analysis (using the PANalytical Axios Advanced XRF spectrometer) to determine their bulk composition. Each of these samples was derived from different parts of the main rock to test heterogeneity of the samples. Table 1 shows the average concentration of each element in the samples. There was significant variation in the concentration of trace elements between the different samples of the analogues which has implications in the uncertainties imposed on the measurements. Since the bulk composition of the analogues is being studied, an average composition estimated from the two regions of the analogue was deemed to be sufficient

Table 1
XRF compositional analysis of the Martian analogues.

Element	Colombian River Basalt		Theo's flow		Mount Erebus	
	WF (%)	Error	WF (%)	Error	WF (%)	Error
Si	23.55	0.12	23.56	0.21	20.40	0.10
Ti	1.78	0.03	0.43	0.01	2.22	0.05
Al	6.92	0.01	2.68	0.08	7.59	0.19
Fe	10.94	0.07	7.80	0.02	8.65	0.13
Mn	0.16	0.01	0.15	0.01	0.15	0.05
Mg	2.54	0.01	8.78	0.28	5.43	0.72
Ca	6.04	0.02	11.49	0.18	6.97	0.22
Na	2.06	0.08	0.67	0.01	3.20	0.05
K	1.11	0.08	0.03	0.01	1.75	0.09
P	0.25	0.08	0.02	0.01	0.36	0.01
S	0.02	0.02	0.02	0.01	0.02	0.01
O	44.34	0.16	43.65	0.38	43.60	0.06
	ppm	Error	ppm	Error	ppm	Error
Ba	503.44	63.15	11.90	7.10	504.09	29.05
Ce	57.81	0.05	9.47	1.24	169.46	0.05
Co	28.80	16.02	59.79	0.05	29.82	19.39
Cr	32.85	24.94	1402.43	29.73	395.35	156.29
Cs	–	–	–	–	3.79	0.05
Cu	20.92	8.87	14.92	0.91	34.25	8.08
Ga	22.86	0.15	8.66	0.73	21.80	1.81
La	24.08	0.05	4.37	0.05	84.46	0.05
Mo	2.45	0.05	1.20	0.15	6.61	0.58
Nb	14.42	0.50	1.73	0.39	117.40	9.25
Nd	28.88	0.05	8.64	0.05	74.05	0.05
Ni	5.96	1.09	278.57	4.67	152.57	46.54
Pb	7.56	0.39	2.48	0.84	2.07	0.39
Rb	33.86	1.44	2.38	0.75	44.29	3.70
Sc	56.92	12.5	67.83	5.83	377.28	6.43
Se	2.91	0.05	1.89	0.05	4.20	0.05
Sn	–	–	–	–	1.56	0.05
Sr	321.24	2.08	45.71	2.19	1009.01	30.57
Th	4.45	0.44	1.35	0.24	6.89	0.8
U	1.99	0.01	1.76	0.05	1.38	0.11
V	400.54	26.21	219.96	8.77	252.48	2.56
Y	45.94	0.95	15.10	0.35	36.69	0.70
Zn	137.86	1.14	65.44	1.87	103.58	2.82
Zr	203.53	1.83	41.83	0.70	483.17	22.43

to give a representative composition measurement of the bulk of the analogue.

In addition to the three terrestrial analogue rocks, several calibration samples were produced at the University of Leicester containing particular elements of interest. Four samples were created; acid washed sand was used as a dilution agent and elements of interest in Martian surface studies (Cl, H and Fe) were added. Boynton et al. [7] have determined the concentration of these elements over the Martian surface based on the γ-ray data collected by the MO γ-ray spectrometer. One sample containing only sand was used as a control. Lithium hydroxide (LiOH) and polyvinylidene chloride (PVDC) were used to simulate a H content of ~0.5% in a planetary surface. The PVDC also contained Cl, although the concentration of Cl would be far in excess of what would be detected on a planetary surface (~9.7%, whereas the concentration on the Martian surface is ~0.5% [7]). Iron filings were used to simulate a silicate-based surface composed of ~12.5% Fe. The combination of using PGAA on the calibration samples and rock analogues had the additional advantage of being able to verify the model for regolith material with a density of ~1–2 g cm^{-3} and more rocky/granular material with a density of ~3 g cm^{-3} that may be encountered on a planetary surface.

Any material used to suspend the calibration samples and analogues would also be exposed to the neutron beam and would contribute to the γ-ray emission. The solution was to use plastic (low-density polyethylene) bags that are composed of predominantly H and C. This would mean that a H peak would be detected from the plastic bag (C has a very small thermal neutron capture cross-sections, so it was unlikely a peak would be detected since the plastic bag had a low mass compared to the calibration samples and analogues).

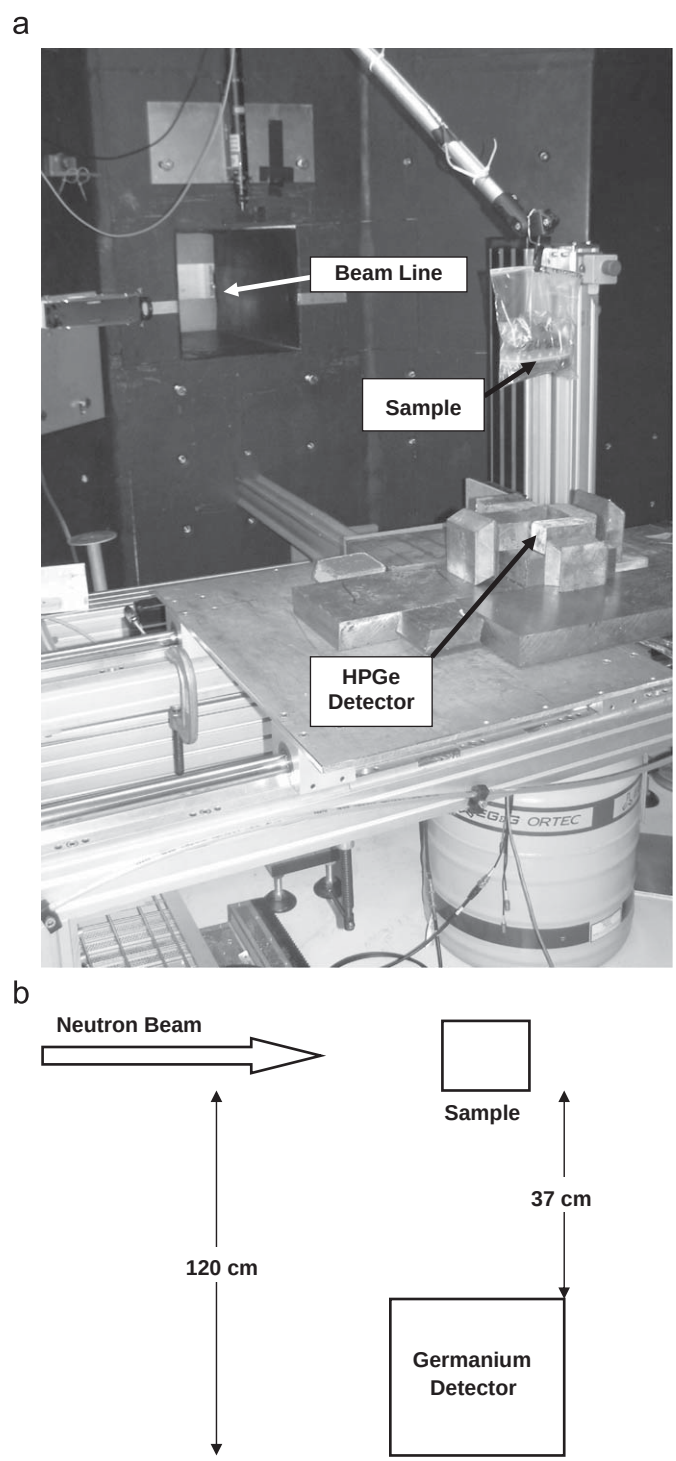

Fig. 2. The experimental set up in the ANTARES chamber.

4. ANTARES test campaign

4.1. Test campaign

A picture of the experimental chamber and the equipment configuration is shown in Figs. 2a and b. The calibration samples and analogues were suspended from a boom such that it was directly in the path of the neutron beam and directly above the γ-ray detector. This configuration was chosen to minimise the activation of the detector by the neutron beam. A large volume Ortec® GEM-15200-P HPGe detector, coupled to an Ortec® 570 spectroscopy amplifier and an Ortec® Maestro-32 MCA emulator (Version 6.06) was used to measure the γ-ray emission from the analogues. The bias was supplied by an Ortec® 459 high-voltage supply.

The detector system response limited the detection of γ-rays to below 4.3 MeV. This meant that important neutron capture γ-ray peaks could not be detected, such as the 7.631 and 7.645 MeV peaks for ^{56}Fe. However, many of the elements with these high energy peaks have lower energy counterparts that could be used for the γ-ray analysis. The HPGe detector was calibrated with several standards; ^{22}Na, ^{54}Mn, ^{88}Y and ^{57}Co.

Three filters were used in the beam line to reduce the dead time of the detector from 100% to 6%. They consisted of:

- a polycrystalline bismuth filter and a lead filter that attenuated the γ-ray background in the beam;
- a 2 mm pinhole aperture that reduced the flux of the neutron beam.

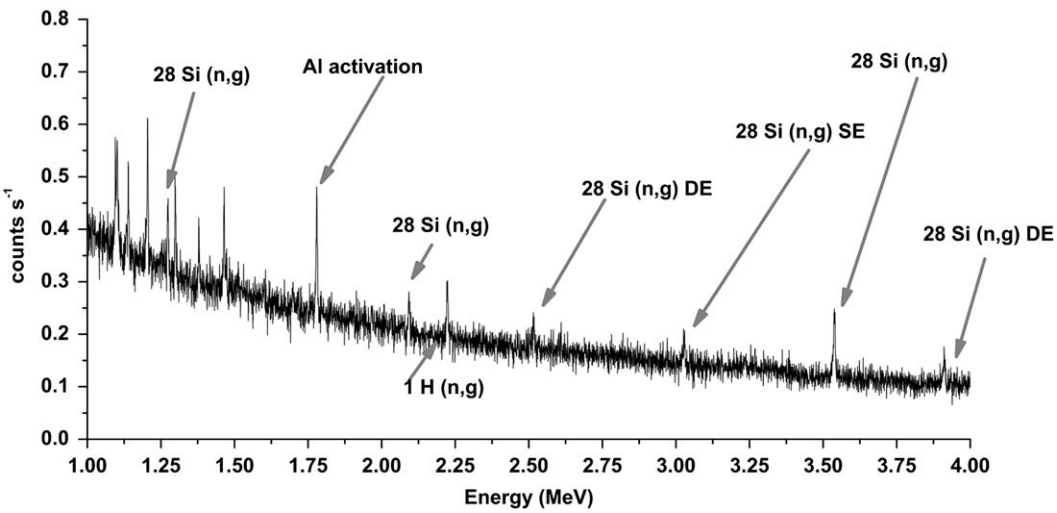

Fig. 3. A portion of the spectrum from the sand calibration sample. The Si, Al and H peaks have been labelled. A single escape peak is labelled as SE and a double escape peak is labelled as DE.

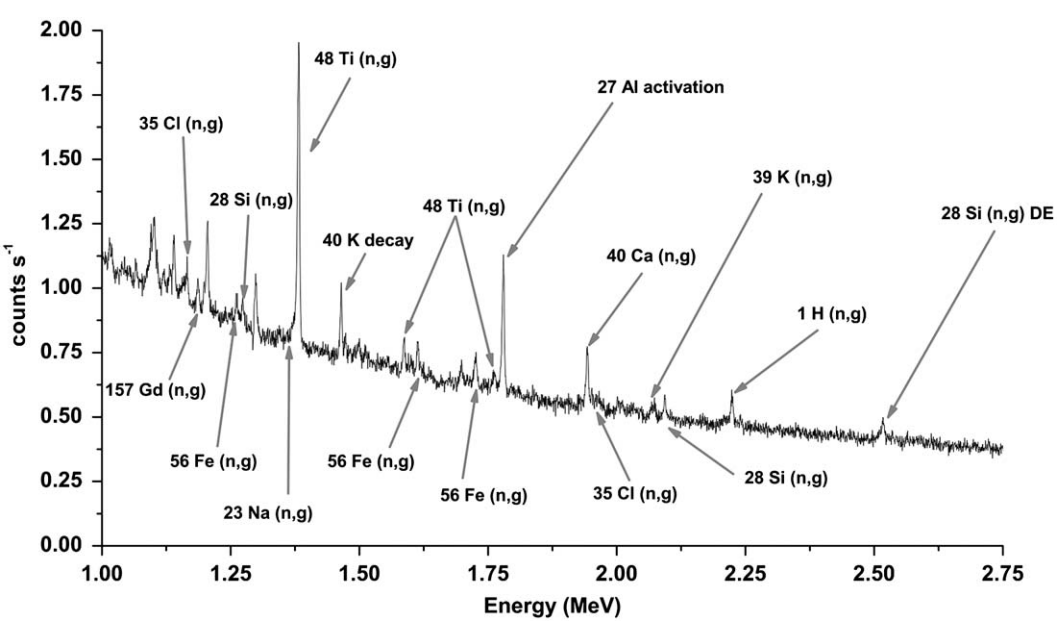

Fig. 4. A portion of the spectrum from the ME analogue with the strongest γ-ray peaks labelled. A single escape peak is labelled as SE and a double escape peak is labelled as DE.

The samples and analogues were exposed to the beam for periods not exceeding 40 min.

4.2. Spectral fitting

A sample γ-ray spectrum from the sand calibration sample is shown in Fig. 3 with the key Si and H (from the sample container bag) γ-ray peaks. For comparison, a spectrum from the ME analogue (the analogue that produced one of the most complex spectra) is shown in Fig. 4. The Al peak was very large compared with the other peaks that were from the calibration sample/analogue. The HPGe detector was supported by an Al structure. The γ-rays from the activation of the Al and the detector contributed to the γ-ray spectrum collected. It was impossible to separate the contributions to the Al γ-ray peaks from the structure and the analogues, therefore the Al peaks could not be used in the verification process. The data was reduced by

(a) subtracting the background that was present with no sample in the beam (the background spectrum was collected over the same time period as the calibration samples/analogues);
(b) fitting and subtracting the additional background created when the sample was in the beam (the sample caused scattering of neutrons from the beam into the material in the chamber). As seen in Figs. 3 and 4, the background from the ME analogue is larger than the sand calibration sample. This is because the ME analogue density was greater than the sand sample, and the ME analogue had a larger surface area exposed to the neutron beam. This caused a greater degree of scatter from the neutron beam as it hit the analogue; the additional scatter increased the γ-ray background from the chamber/Al structure/detector in the ME spectrum;
(c) identifying each peak and calculating the peak areas (the count rate of each peak).

It was possible to identify many lines in the spectra that did not originate from the calibration samples and analogues. Although the beam was well collimated, thermal neutron scatter contributed to the signal by interacting with other structures in the experimental facility. Many of the γ-ray background peaks could be attributed to the HPGe detector itself. The background signal from the detector changed once a sample was placed in the beam line and there was an increase in the intensity of the peaks associated with the HPGe crystal. This was attributed to increased neutron scatter. For example, the HPGe neutron capture line at 0.595 MeV from the HPGe detector approximately doubled in intensity when the ME analogue was placed in the beam. The background from the experimental chamber also increased once a sample was placed in the beam. The B neutron capture reaction $^{10}B(n,\alpha\gamma)^{7}Li$ from the borated polyethylene lining the chamber walls produced a broad peak at 0.478 MeV that also doubled in intensity when the ME analogue was placed in the beam.

The remaining peaks in the spectra were identified using the prompt γ-ray spectrum catalogue [19]. The γ-ray libraries of the major elements present in the samples were examined to identify each of the γ-ray peaks, since these had the greatest abundances.

4.3. Results

The γ-ray peak element identification, cross-section and energy of each peak detected is shown in Table 2. The γ-ray peaks from the background have been omitted. Revay et al. [19] was used to identify the γ-ray lines and any interference peaks.

Table 3 shows the average count rate per gram of each of the statistically significant peaks in the analogues and calibration samples that could be used in elemental analysis (the peaks highlighted in bold in Table 2). Only the peaks with less than a 20% uncertainty in the count rate are included. This meant that only the major elements of the analogues could be determined because they were present in large enough quantities to give

Table 2
γ-rays detected in the calibration samples and analogues by the HPGe detector at the ANTARES facility [19].

Energy (MeV)	Element and reaction	Cross-section (barns)
0.292	^{35}Cl (n,g)	0.089
0.314	^{55}Mn (n,g)	1.46
0.342	^{48}Ti (n,g)	1.84
0.352	**^{56}Fe (n,g)**	**0.273**
0.390	^{24}Mg (n,g)	0.00586
0.436	^{35}Cl (n,g)	0.309
0.520	^{35}Cl (n,g)	7.58
0.585	^{24}Mg (n,g)	0.0314
0.632	^{35}Cl (n,g)	0.111
0.692	^{56}Fe (n,g)	0.137
0.768	**^{39}K (n,g)**	**0.903**
0.787	^{35}Cl (n,g)/^{35}Cl (n,g)	171/271
0.847	^{55}Mn (n,g)/^{56}Fe (IN)	13.1
0.898	^{56}Fe (n,g)	0.054
0.936	^{35}Cl (n,g)	0.172
0.984	^{48}Ti (n,g)	0.114
1.131	^{35}Cl (n,g)	0.626
1.165	**^{35}Cl (n,g)**	**8.91**
1.186	^{39}K (n,g)	0.16
1.260	^{56}Fe (n,g)	0.0684
1.273	**^{28}Si (n,g)**	**0.0289**
1.374	^{23}Na (n,g)[a]	0.53
1.382	**^{48}Ti (n,g)**	**5.18**
1.461	^{40}K decay	
1.499	^{48}Ti (n,g)	0.297
1.586	**^{48}Ti (n,g)**	**0.624**
1.601	^{35}Cl (n,g)	1.21
1.613/1.614/1.619	^{56}Fe (n,g)/^{39}K (n,g)/^{39}K (n,g)	0.153/0.119/0.13
1.726	**^{56}Fe (n,g)**	**0.181**
1.762	^{48}Ti (n,g)	0.311
1.943	**^{40}Ca (n,g)**	**0.352**
1.951	**^{35}Cl (n,g)**	**6.33**
1.960	**^{35}Cl (n,g)**	**4.10**
2.001	**^{40}Ca (n,g)**	**0.0659**
2.034	^{35}Cl (n,g)	0.239
2.074	^{39}K (n,g)	0.137
2.075	^{35}Cl (n,g)	0.252
2.094	^{28}Si (n,g)	0.0331
2.209	^{23}Na (n,g)	0.0259
2.224	**^{1}H (n,g)**	**0.333**
2.311	^{35}Cl (n,g)	0.35
2.519	^{23}Na (n,g)	0.0699
2.677	^{35}Cl (n,g)	0.533
2.801	^{35}Cl (n,g)	0.183
2.846	^{35}Cl (n,g)	0.349
2.870	^{35}Cl (n,g)	1.82
2.975	^{35}Cl (n,g)	0.377
3.026/3.027	^{48}Ti (n,g)/^{48}Ti (n,g)	1 3/26
3.061	^{35}Cl (n,g)	1.13
3.540	**^{28}Si (n,g)**	**0.119**
3.916	^{24}Mg (n,g)/^{48}Ti (n,g)/^{48}Ti (n,g)	0.032/0.13/0.0839

IN = inelastic scatter reaction. The escape peaks created in the HPGe detector are not included in this table. The elemental peaks found to be suitable to estimate the composition of a planetary surface within the Monte Carlo model (i.e. the experimental to modelled ratios were close to 1) are highlighted in bold. These peaks were chosen because they have no interference peaks from other elements in the analogues and the γ-ray emission rates were relatively high compared to other γ-rays emitted by these elements. The Mg peaks either had overlaps with other element peaks or were not statistically significant peaks (i.e. had uncertainties >20%). The Na peaks were either not statistically significant or were decay peaks from the decay of ^{24}Na (created by neutron capture); decays from radioactivity were not modelled.

[a] Indicates that the isotope created by the capture is radioactive and that γ-ray line is due to the radioactive decay of the isotope.

Table 3
The average count rate per gram of each element within the analogues/calibration samples for the primary elemental peaks (those highlighted in Table 2).

Energy (MeV)	Element	Average count rate per gram (counts $s^{-1} g^{-1}$)	Error	Present in analogue/calibration sample
0.342	^{48}Ti	0.2578	2.50E−03	ME, CRB
0.768	^{39}K	0.0839	7.41E−03	ME, CRB
1.164	^{35}Cl	0.5445	1.59E−02	ME, PVDC
1.382	^{48}Ti	0.2338	3.17E−02	ME, TF, CRB
1.586	^{48}Ti	0.0285	7.67E−04	ME
1.726	^{56}Fe	0.0067	1.22E−03	ME, TF, CRB, FE
1.943	^{40}Ca	0.0185	1.51E−03	ME, TF, CRB
1.951	^{35}Cl	0.2508	2.31E−02	ME, PVDC
1.960	^{35}Cl	0.1357	2.19E−04	PVDC
2.001	^{40}Ca	0.0032	6.26E−04	ME, TF
2.094	^{28}Si	0.0019	3.37E−04	ME, TF, CRB, SI, FE, LiOH, PVDC
2.224	^{1}H	0.6081	4.35E−02	ME, TF, CRB, SI, FE, LiOH, PVDC
2.677	^{35}Cl	0.0133	7.02E−04	PVDC
2.870	^{35}Cl	0.0517	3.55E−04	PVDC
3.540	^{28}Si	0.0038	5.43E−04	ME, TF, CRB, SI, FE, LiOH, PVDC

ME = Mount Erebus; TF = Theo's Flow; CRB = Colombian River Basalt; SI = sand calibration source; FE = Fe calibration source; PVDC = Cl and H calibration source; LiOH = H calibration source.

significant peaks. Very few statistically significant peaks could be attributed to trace elements due to the limited exposure times of each analogue/sample and the high γ-ray background. The trace element peaks detected typically had large neutron capture cross-sections, e.g. the capture cross-sections for Cl are several barns or more for the peaks listed in Table 2.

The ME analogue had γ-ray peaks at 1.165 and 1.951 MeV, which could not be attributed to any of the strong γ-ray peaks of the major elements. These γ-ray energies were consistent with the strongest γ-ray lines of Cl (1.165 and 1.951 MeV). The concentration of this element was not measured in the XRF analysis carried out on the analogue. Although Cl would only be present in trace amounts (ppm), its neutron capture cross-section is large (see Table 2) and there is evidence that ME and other volcanic basalts may contain Cl [20–22]. The measured count rate from the Cl activation in the ME analogue was consistent with the count rate per gram of Cl in the sand and PVDC analogue, indicating that Cl was present at a level of 1080 ± 100 ppm. The presence of Cl can be verified when modelling the ME γ-ray emission in MCNPX.

By comparing the count rate per gram of H present in the analogues/samples, it was evident that the TF analogue contained a small amount of H (possibly in the form of a hydride), estimated at $\sim 1.2 \pm 0.1$ g or $0.17 \pm 0.015\%$ of the total mass of the TF analogue [21].

5. Modelling the γ-ray emission using MCNPX

As described in Section 1.3, MCNPX was used to model the γ-ray emissions from the calibration samples and analogues. A model of the experiment was created with the geometry shown in Fig. 2b. Each calibration sample/analogue's bulk density was assumed to be uniform throughout the sample/analogue (the density of the calibration samples was approximately $\sim 1.6\,g\,cm^{-3}$ and the analogues $3\,g\,cm^{-3}$). The background from the chamber and the detector (without any calibration sample/analogue in the beam) was measured and subtracted so this was not considered in the model (this reduces the complexity of the model).

5.1. The model

The neutron input spectrum of the model covered cold and thermal energies, so the focus of the modelling was on neutron capture reactions. The thermal neutron spectrum [17] was divided into 14 bins of increasing energy. The average contribution to the total neutron flux from each bin formed the input into the MCNPX program; Table 4 lists the average energy of the bins and the contribution to the neutron flux for each energy bin. The total thermal neutron flux was measured to be $\sim 1 \times 10^{7}\,cm^{-2}\,s^{-1}$ [17].

Table 4
The neutron spectrum divided into 14 bins to input into the MCNPX program.

Energy bin (median value (MeV))	Fraction of the neutron flux
6.18E−10	0.072
1.13E−09	0.024
1.45E−09	0.034
1.94E−09	0.053
2.70E−09	0.077
3.62E−09	0.050
4.53E−09	0.053
5.81E−09	0.065
7.74E−09	0.093
1.08E−08	0.109
1.62E−08	0.115
2.67E−08	0.115
5.23E−08	0.094
1.45E−07	0.046

A tally to track the number of γ-rays entering the void detector was selected for the simulation. Only γ-rays that entered the detector cell volume were tallied. The detector was modelled as a cylinder of dimensions $\varnothing$ 4.79 cm × 5.04 cm and was positioned 37 cm below the analogues as shown in Fig. 2b. Only neutron capture reactions were considered, i.e. all other secondary processes including Bremsstrahlung reactions were switched off. The calibration samples and analogues were modelled with the compositions given in Section 3 and in Table 1. In addition, 1080 ppm of Cl was added to the simulated composition of the ME analogue (see Section 4.3). Based on the experimental results obtained from the comparison with the PVDC calibration sample, 1.2 g of H was added to the composition of the TF analogue, as described in Section 4.3. Based on the literature [20,21,23], 4 ppm of Sm and Gd were added to the ME and CRB analogue compositions in the model. Sm and Gd are strong thermal neutron absorbers and their presence in the analogues could affect the resultant γ-ray emission.

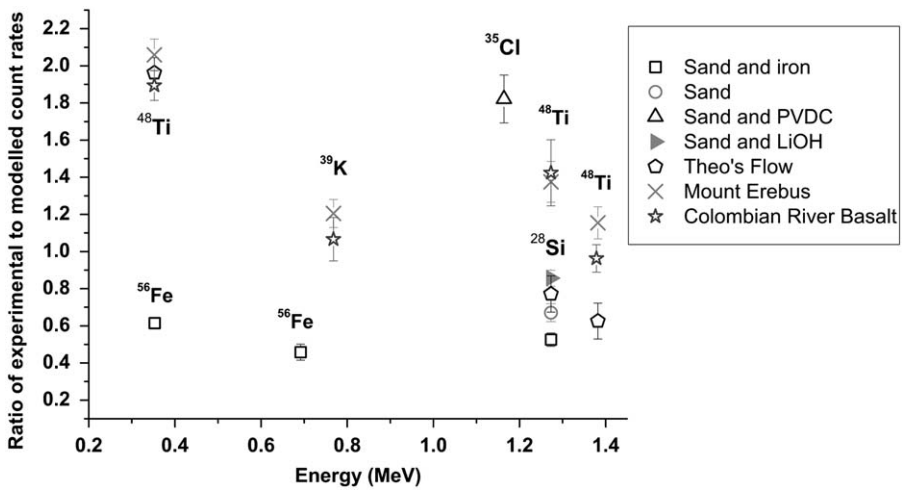

Fig. 5. The ratio of the experimental to the modelled count rates for each γ-ray peak present in the calibration samples and analogues that had less than a 20% uncertainty in the energy range 0.2–1.45 MeV and are suitable to use to determine the concentration of a particular element.

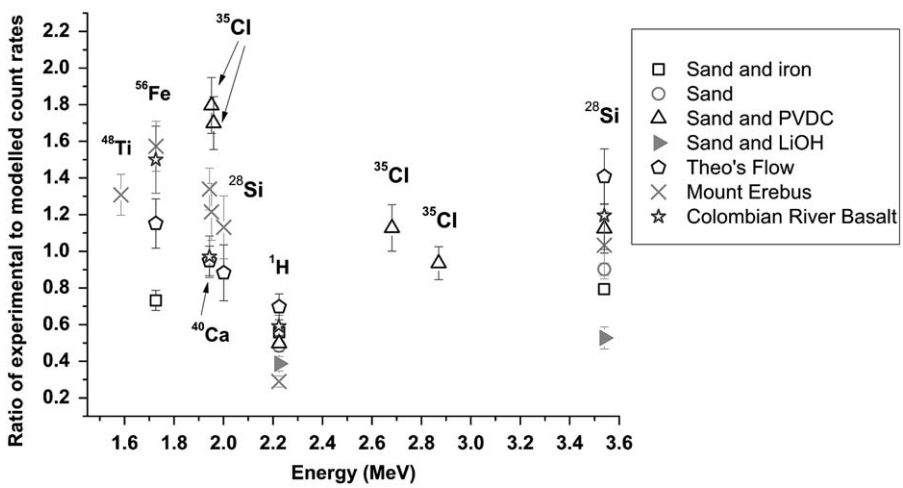

Fig. 6. The ratio of the experimental to the modelled count rates for each γ-ray peak present in the calibration samples and analogues that had less than a 20% uncertainty in the energy range 1.45–4 MeV and are suitable to use to determine the concentration of a particular element.

5.2. MCNPX cross-sectional information

MCNPX combines cross-section and reaction libraries with physical models in a single code [12]. The neutron cross-sectional data and photon production data used were from the Evaluated Nuclear Data File (ENDF/B-VI and -VII) libraries. Only low-energy neutrons were modelled, so the data up to neutron energies of 20 MeV were used where available. Elements with no cross-sectional information in the ENDF databases or that had no photon production data utilised models by default. Photon production cross-sections do not exist for Zr so no γ-ray data could be generated for that element. The H signal from the plastic bags used to suspend the analogues was modelled separately.

6. Results

The γ-ray count rates given by the model were converted into the count rates detected by the HPGe detector using a response function ε as a function of energy E that was calculated in Skidmore et al. [11], corrected for the size of the detector and is expressed as $\varepsilon(E) = kE(\text{keV})^{-0.85\pm0.03}$ where k is a constant. The ratios of the experimental to modelled results are shown in Figs. 5 and 6 for elemental peaks that had no interferences with other elemental peaks and were statistically significant ($<$20% uncertainty). The modelled data is in good agreement with the experimental data; the experimental and modelled count rates do not differ by more than a factor of two; the possible reasons for the discrepancy are detailed below.

7. Discussion—comparison between model and experimental results

In general the modelled and experimental data were within a factor of two of each other. However, Figs. 5 and 6 show that some of the experimental data to model ratios had values <1. There are several possible reasons for this, the primary one being that the model did not take into account that the presence of the sample/analogue in the beam changed the background count rate in the γ-ray spectrum. γ-rays from the chamber and the HPGe detector

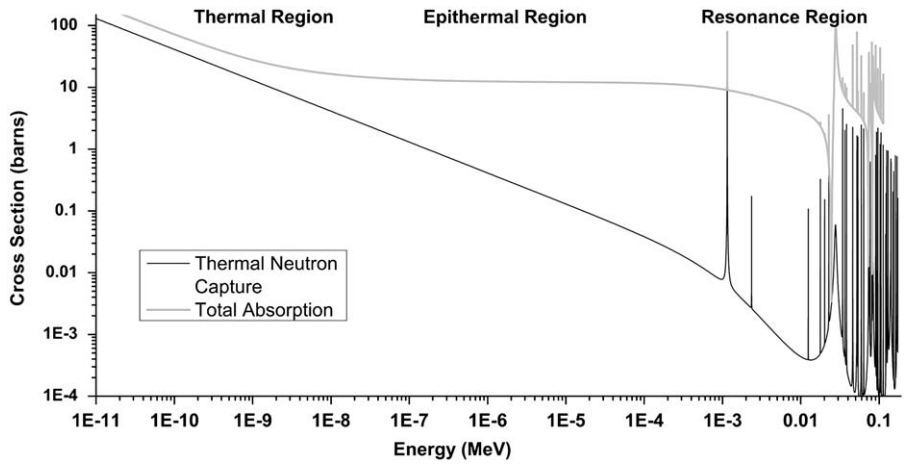

Fig. 7. The total neutron absorption and neutron capture cross-section with energy for Fe [24].

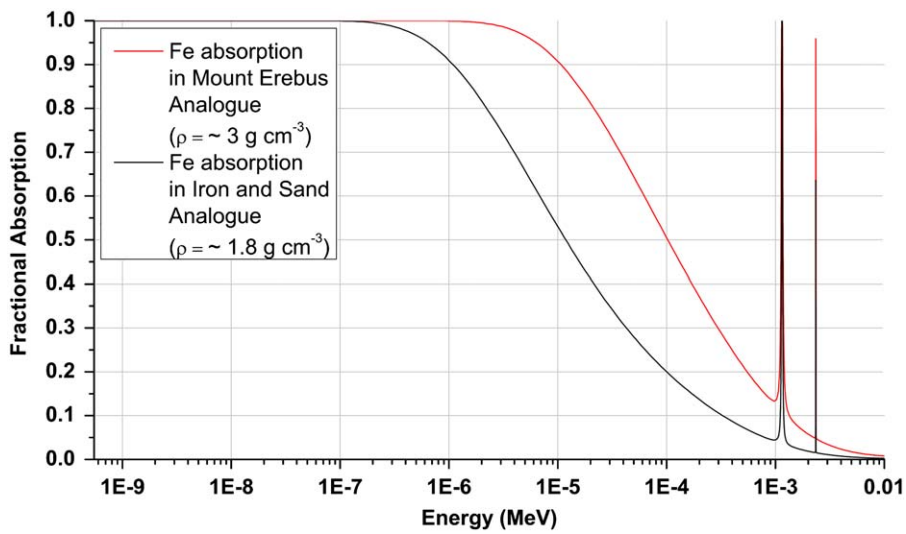

Fig. 8. The fractional absorption due to capture of neutrons with energy by Fe in the ME analogue and Fe calibration samples [24]. As seen in the diagram the larger density of the ME analogue means that the Fe present in it would capture a larger portion of higher energy neutrons than the lower density calibration sample. Neutrons in the epithermal region ($>10^{-6}$ MeV) were not modelled; this may be the cause of the discrepancy between the Fe γ-ray peak count rates in the modelled and experimental data.

could have reduced the signal to background ratio. This effect could be partly responsible for suppressing the experimental count rate. The only way to account for the additional contributions from scatter is to include the chamber and its contents in the simulation, in particular any structures close to the detector.

In the analogues the ratios between the experimental and modelled data were generally >1 for elements such as Cl and Fe. There were several reasons for this:

- The analogues were assumed to be homogeneous in the model. The variation in composition measured by sampling different parts of the analogues highlights that this was not the case.
- The analogues were roughly shaped, i.e. their thicknesses varied across their surfaces, whereas in the MCNPX model they were modelled as cuboids, therefore there was varying absorption across the samples/analogues.
- The neutron spectrum was separated into coarse energy bins and the epithermal neutron content of the beam was not modelled. The epithermal content of the ANTARES beam was not measured by Lorenz et al. [17], therefore the epithermal neutron flux could not be accurately modelled. Fig. 7 shows the total neutron cross-section and neutron capture cross-sections as a function of energy for Fe. There are a number of resonances in the cross-section at epithermal energies. The effect of these resonances was not taken into account and may have caused additional capture reactions, causing the ratio of the experimental to modelled count rates to be higher. For example in Fig. 7 there is a resonance region between 0.001 and 0.1 MeV which increases the neutron capture cross-section at resonance energies from $\sim$ 0.01 to $\sim$ 10 b, approximately the same cross-section as that at thermal energies.

Only when Cl was present in the calibration samples a ratio of >1 was measured. The count rates of the 1.164, 1.951 and 1.960 MeV Cl lines in the experimental spectra were far higher than in the modelled spectra. The ratio of the modelled to experimental data was $\sim$2, whereas for the other Cl peaks (e.g. 1.165 MeV) this was not the case. Possible explanations of this are described in Section 7.1.

7.1. Limitations of the Monte Carlo model

Although the experimental and modelled data for the Fe peaks in the Fe calibration sample were in good agreement, the ratios for the experimental to modelled Fe peaks in the ME, CRB and TF analogues were well in excess of 1 (the experimental Fe count rate was far higher than in the model). However, the three analogues with similar densities of $\sim 3\,\mathrm{g\,cm^{-3}}$, have experimental to modelled ratios which are comparable within error. The reason for this may be due to the uncertainty in the epithermal neutron content of the beam, coupled with the higher density of the analogues ($\sim 3\,\mathrm{g\,cm^{-3}}$) compared to the calibration sample ($\sim 1.8\,\mathrm{g\,cm^{-3}}$). The denser the analogue in the neutron beam, the greater the absorption of the thermal and epithermal content of the neutron beam. Fig. 8 shows the fractional capture of neutrons with varying energy when the density of the material containing the Fe is ~ 1.8 and $\sim 3\,\mathrm{g\,cm^{-3}}$, respectively. The epithermal component of the beam would contribute to neutron capture reactions in the denser material. This may also explain why the experimental to modelled count rate ratio was >1 for the Cl peaks in the sand and PVDC calibration sample. Cl is a good absorber of neutrons across thermal and epithermal energies (see Table 3 and Ref. [24]). In a model simulating the GCR interaction with a planetary surface this discrepancy should not arise given that the complete neutron spectrum would be modelled. In the case of the GCR-based model the neutrons are secondary particles produced by proton interactions.

One of the disadvantages of this experiment is that it was not possible to examine the performance of the model at higher energies (4–10 MeV), where many important γ-rays may be found e.g. ^{56}Fe at 7.631 and 7.645 MeV. These limitations were imposed by the detector system response. A broader response could be achieved in the future by replacing the detector preamplifier with one that could process very high count rates (reducing the dead time), replacing the amplifier with one that could provide the necessary gain to process γ-rays up to 10 MeV, and using a detector that has a larger volume. However, the model needs to be verified at both higher γ-ray energies (4–10 MeV) and over a broader neutron spectral range (neutrons of thermal, epithermal and fast energies). It is also desirable to compare the model to real data gathered by γ-ray spectrometers in orbit around a planetary body. In order to achieve this data from LP and MO have been used to verify the model. The response of γ-ray spectrometers on these spacecraft range up to $\sim$10 MeV. This work will be reported in a future publication. The results from these studies could be implemented in determining the performance of the proposed geophysical instrument [11] for composition, heat flow and dating applications.

Another limitation was the inability to detect the trace elements within the analogues in the experimental campaign, with the exception of Cl. The concentration of the trace elements was low compared to the major elements; therefore the background may have been a limiting factor. The peaks detected that could be attributed to trace elements had large uncertainties in both the content and the count rate. A possible solution to these problems would have been to expose the analogues for longer time periods (this may be possible in the future) and to sample more regions of the analogues to reduce the uncertainty in the content of the analogue. The ME analogue model was used to estimate the time required to get a 10% precision on some of the trace element peaks if a future study was carried out; the strongest peaks of Cr and Ba [19] could get a 10% precision within an exposure time of a few hours; however, for elements that have concentrations of only a few ppm (e.g. Th, U), the time required is >30 h. The Monte Carlo model will be used to determine the limits of detectability of trace elements on a planetary surface for various detectors given a realistic measurement period for a typical in-situ profile and will be reported in future publications. The Mg peaks at 0.390 and 0.585 MeV and the radioactive decay peaks of Na (created by neutron capture) were not present in the results from the model so the experimental peaks could not be used in the model verification.

8. Conclusion

Several Martian analogues and calibration samples of known composition were exposed to a thermal neutron beam at FRM II in Munich, Germany to induce γ-ray emission. It was found that the γ-ray emission could be attributed to the major rock-forming elements of the analogues and the elements of the calibration samples. The γ-ray information also indicated the presence of Cl in the ME analogue, an element not discovered in the original XRF analysis and an estimate of the concentration of Cl present was obtained. The γ-ray emissions and the compositions were used to verify a Monte Carlo model being developed by Skidmore et al. The ratio between the modelled and experimental count rates gave consistent results for elements that were common to the analogues and calibration samples with the exception of Fe; however, this has been attributed to the different densities of the iron and sand calibration sample and the analogues. The model had a number of limitations, which will be remedied by verifying the model using data acquired by the γ-ray spectrometers on 2001 Mars Odyssey and Lunar Prospector. This work will be reported in future publications.

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